

**Amrita Vishwa Vidyapeettam**  
**Amrita School of Engineering Bengaluru Campus**  
**Department of Mathematics**

**19MAT205 – Probability and Random Processes**

**ASSIGNMENT QUESTIONS**

1. What is the probability of getting 53 Sundays in a leap year?
2. Choose a number at random from 1 to 5.
  - (i) What is the probability of each outcome?
  - (ii) What is the probability that the number chosen is even?
  - (iii) What is the probability that the number chosen is odd?
  - (iv) Is the sum of probabilities in (ii) and (iii) one? If yes, why so?
3. Are the following statements true or false? If false, explain why it is false.
  - (i) The probability that a mineral sample will contain silver is 0.38 and the probability that it will not contain silver is 0.52.
  - (ii) The probability that a drilling operation will be a success is 0.34 and the probability that it will not be a success is  $-0.66$ .
4. A lottery sells tickets numbered from 00001 through 50000. What is the probability of drawing a number that is divisible by 200
5. A campus police department needs new bicycles for its patrol persons and the probabilities are 0.17, 0.22, 0.03, 0.29, 0.21 and 0.08 that it will buy Bianshe, Cannonhill, Fishin, Giante, Trec, or H.T. Find the probability that the department buys
  - (i) Cannonhill or Trec
  - (ii) Bianshe, Giante or Trec
  - (iii) anything other than Fishin or HT
6. When we roll a pair of balanced dice, what are the probabilities of getting
  - (i) 5    (ii) 10    (iii) 5 or 10    (iv) 2 or 12    (v) a sum greater than 3    (vi) an even sum
7. A glass jar contains 6 red, 5 green, 8 blue and 3 yellow marbles. If two marbles are taken at random from the jar with replacement, what is the probability of choosing
  - (i) a red marble and a yellow marble?
  - (ii) 2 green marbles ?
  - (iii) a blue marble and a red marble ?
  - (iv) 2 yellow marbles?
8. Find the corresponding probabilities in the previous problem with the assumption that marbles are taken without replacement ?
9. The supervisor of a group of 20 construction workers wants to get the opinion of two of them (to be selected at random) about certain new safety regulations. If 12 workers favour the new regulations and the other 8 are against them, what is the probability that both of the workers chosen by the supervisor will be against the new safety regulations?
10. The probability that a new airport will get an award for its design is 0.16, the probability that it will get an award for the efficient use of materials is 0.24, and the probability that it will get both awards is 0.11
  - (i) What is the probability that it will get atleast one of the two awards?
  - (ii) What is the probability that it will get only one of the two awards?
  - (iii) Find the probability that the airport will get the design award given that it got the award for efficient use of materials.
  - (iv) Find the probability that the airport will get the award for efficient use of materials given that it didn't get the award for design.

11. The following table gives the number of vehicles in a showroom categorized in terms of performance and looks.

|              |           | <u>Performance</u> |      |      |
|--------------|-----------|--------------------|------|------|
|              |           | Excellent          | Good | Poor |
| <u>Looks</u> | Excellent | 45                 | 32   | 1    |
|              | Good      | 58                 | 7    | 2    |
|              | Poor      | 12                 | 3    | 0    |

If A represents the event of selecting a vehicle with excellent performance and B represents the event of selecting a vehicle with excellent looks, find the following probabilities:  $P(A)$ ,  $P(B)$ ,  $P(A \cap B)$ ,  $P(A^c)$ ,  $P(A \cup B)$ ,  $P(A^c \cap B^c)$ ,  $P(A/B)$ ,  $P(A/B^c)$ .

12. The probability that an integrated circuit chip will have defective etching is 0.12; the probability that it will have a crack defect is 0.29, and the probability that it has both defective is 0.07
- What is the probability that a newly manufactured chip will have either an etching or a crack defect.
  - What is the probability that a newly manufactured chip will have neither defect?
  - What is the probability that has defective etching will also have a crack defect?
  - What is the probability that has a crack defect will also have a defective etching?
13. A coin is tossed and a single 6-sided die is rolled. Find the probability of landing on the head side of the coin and rolling a 3 on the die.
14. A card is chosen at random from a deck of 52 cards. It is then replaced and a second card is chosen. What is the probability of choosing a jack and an eight?
15. Spin a spinner numbered 1 to 7, and toss a coin. What is the probability of getting an odd number on the spinner and a tail on the coin?
16. If  $P(A)=0.8$ ,  $P(B^c)=0.65$  and  $P(A \cap B)=0.28$ , are the events A and B independent?
17. Suppose that  $P(A/B) = 0.2$ ,  $P(A/B^c) = 0.3$  and  $P(B) = 0.8$ , What is  $P(A)$ ?
18. Find the probabilities of getting
- 8 heads in a row with a balanced coin.
  - Three 3s and then a 4 or a 5 in 4 rolls of a balanced die.
  - 5 multiple-choice questions answered correctly, if for each question the probability of answering it correctly is  $1/3$ .
19. Suppose 2% of cotton fabric rolls and 3% of nylon fabric rolls contain flaws. Of the rolls used by a manufacturer, 70% are cotton and 30% are nylon. What is the probability that a randomly selected roll used by the manufacturer contains flaws?
20. The edge roughness of slit paper products increases as knife blades wear. Only 1% of the products slit with the new blades have rough edges, 3% of products slit with blades of average sharpness exhibit roughness, and 5% of products slit with worn blades exhibit roughness. If 25% of the blades in manufacturing are new, 60% are of average sharpness and 15% are worn, what is the proportion of products that exhibit edge roughness?
21. A certain firm has plants A, B and C producing, respectively, 35%, 15% and 50%, of the total out. The probabilities of a non-defective product are, respectively, 0.75, 0.95 and 0.85. If a consumer receives a defective product, what is the probability that it came from plant C.

22. A computer center has three printers A, B, C which print at different speeds. Programs are routed to the first available printer. The probability that a program is routed to printers A, B, C are 0.6, 0.3 & 0.1 respectively. Occasionally a printer will jam and destroy a printout. The probability that printer the will jam A, B, C are 0.01, 0.05, 0.04 respectively. Your printout is destroyed when a printer jams. What is the probability that (i) printer A is involved? (ii) Printer B is involved (iii) Printer C is involved?
23. Customers are used to evaluate preliminary product designs. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful and 25% have been poor products.
- What is the probability that a product attains a good review?
  - If a new design attains a good review, what is the probability that it will be a highly successful product?
  - If a product does not attain a good review, what is the probability that it will be highly successful product?
24. Bean seeds from supplier-A have 85% germination rate and those from supplier-B have a 75% germination rate. A seed packing company purchased 40% of the bean seeds from the supplier-A and 60% from supplier-B and mixed these seeds together. Given that a seed germinates, find the probability that the seed was purchased from supplier B.
25. Suppose that  $P(A/B) = 0.7$ ,  $P(A) = 0.5$  and  $P(B) = 0.2$ . Determine  $P(B/A)$ .
26. Suppose that  $P(A/B) = 0.4$ ,  $P(A/B') = 0.2$  and  $P(B) = 0.8$ . Determine  $P(B/A)$ .
27. If  $X$  is a discrete random variable which takes the values  $-2, -1, 0, 1$  and  $2$  and if the corresponding probability mass function is given by :
- |        |       |       |       |       |       |
|--------|-------|-------|-------|-------|-------|
| $x$    | $-2$  | $-1$  | $0$   | $1$   | $2$   |
| $f(x)$ | $1/8$ | $2/8$ | $2/8$ | $2/8$ | $1/8$ |
- Find (a)  $P(X \leq 2)$  (b)  $P(X > -2)$  (c)  $P(-1 \leq X \leq 1)$  (d)  $P(X \leq -1 \text{ or } X = 2)$   
 (e) Cumulative Distribution Function of  $X$
28. If  $f(x) = \frac{2x+1}{25}$ ,  $x=0,1,2,3,4$  Find (a)  $P(X=4)$  (b)  $P(X \leq 1)$  (c)  $P(2 \leq X < 4)$  (d)  $P(X > -10)$   
 (e) Mean and Variance of  $X$  (f) Cumulative Distribution Function of  $X$
29. In a semiconductor manufacturing process, three wafers from a lot are tested. Each wafer is classified as pass or fail. Assume that the probability that a wafer passes the test is 0.8 and that wafers are independent. Determine the probability mass function of the number of wafers from a lot that pass the test. Also obtain the distribution function for the random variable.
30. The range of the random variable  $X$  is  $\{0, 1, 2, 3, k\}$  where  $k$  is unknown. If each value is equally likely and the mean of  $X$  is 6, determine  $k$ .
31. A random variable  $X$  has the following probability distribution.
- |          |       |      |       |      |       |      |
|----------|-------|------|-------|------|-------|------|
| $X$ :    | $-2$  | $-1$ | $0$   | $1$  | $2$   | $3$  |
| $f(X)$ : | $0.1$ | $K$  | $0.2$ | $2K$ | $0.3$ | $3K$ |
- (a) Find  $K$  (b) Evaluate  $P(X < 2)$  and  $P(-2 < X < 2)$  (c) Find the CDF of  $X$  (d) Evaluate the mean of  $X$ .  
 (e) Verify that the values in the table follows PMF.

32. If the probability density function of a random variable is given by

$$f(x) = \begin{cases} kx^3 & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the value of  $k$  and the probability that the random variable takes on a value:

(a) between  $\frac{1}{4}$  and  $\frac{3}{4}$  (b) greater than  $\frac{2}{3}$ .

(b) Also find the corresponding distribution function and use it to determine the probabilities that a random variable having this distribution function will take on a value greater than 0.8.

33. Suppose that  $f(x) = x/8$  for  $3 < x < 5$ . Determine  $P(X < 4)$ ,  $P(X > 3.5)$  and  $P(X < 3.5 \text{ or } X > 4.5)$ .

34. Suppose  $f(x) = 0.25$  for  $0 < x < 4$ . Determine the mean and variance of  $X$ .

35. Suppose  $f(x) = 1.5x^2$  for  $-1 < x < 1$ . Determine the mean and variance of  $X$ .

36. Three major parties A, B, C are contending for power in the election of a state and the chance of their winning the election is in the ratio 1:3:5. The parties A, B, C respectively have probabilities of banning the online lottery  $\frac{2}{3}$ ,  $\frac{1}{3}$ ,  $\frac{3}{5}$ .

(i) What is the probability that there will be a ban on the online lottery in the state?

(ii) What is the probability the ban is from party C?

37. Find the distribution of random variable  $X$  whose PDF is  $f(x) = \begin{cases} 3e^{-3x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$  and use it to find the probability that the random variable will take on a values less than or equal to 1.

38. If the PDF of a Random Variable  $X$  is given by  $f(x) = \begin{cases} x & \text{for } 0 < x < 1 \\ 2 - x & \text{for } 1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$ , find the CDF of  $X$ , mean

and variance of  $X$ . Also find the probabilities that the RV  $X$  will take on a value (a) between 0.2 and 0.8 (b) between 0.6 and 1.2.

39. Given a Random Variable  $X$  with PDF  $f(x) = \frac{k}{1+x^2}$  for  $-\infty < x < \infty$ , find  $k$  and also find  $P(X \geq 0)$  and the find the CDF of  $X$ .

40.  $X$  is a continuous RV with PDF given by  $f(x) = \begin{cases} kx & \text{in } 0 \leq x \leq 2 \\ 2k & \text{in } 2 \leq x \leq 4 \\ 6k - kx & \text{in } 4 \leq x \leq 6 \\ 0 & \text{elsewhere} \end{cases}$ . Find the value of  $k$  and

also the CDF of  $X$ .

41. If the moment generating function for a random variable  $X$  is given by  $M_x(t) = (e^{2t} - e^{-t})$ ,  $t > 0$ , find the mean and variance of  $X$ .

42. The random variable  $X$  has a binomial distribution with  $n=10$  and  $p=0.01$ . Determine:

(a)  $P(X=5)$  (b)  $P(X \leq 2)$  (c)  $P(X \geq 9)$  (d)  $P(3 \leq X < 5)$  (e) Mean and variance of  $X$

43. An electronic product contains 40 integrated circuits. The probability that any integrated circuit is defective is 0.01, and the integrated circuits are independent. The product operates only if there are no defective integrated circuits. What is the probability that the product operates?

44. The phone lines to an airline reservation system are occupied 40% of the time. Assume that the events that the lines are occupied on successive calls are independent. Assume that 10 calls are placed to the airline.
- What is the probability that for exactly three calls the lines are occupied?
  - What is the probability that for at least one call the lines are not occupied?
  - What is the expected number of calls in which the lines are all occupied?
45. Human error is given as a reason for 75% of all accidents in a plant. Use the formula for the binomial distribution to find the probability that human error will be given as the reason for 2 of the next 4 accidents.
46. An agricultural cooperative claims that 90% of the watermelons shipped out are ripe and ready to eat. Find the probabilities that among 18 watermelons shipped out,
- all 18 are ripe and ready to eat.
  - Atleast 16 are ripe and ready to eat.
  - Atmost 14 are ripe and ready to eat
47. Among the 12 solar collectors on display at a trade show, 9 are flat-plate collectors and the others are concentrating collectors. If a person visiting the show randomly selects 4 of the solar collectors to check out, what is the probability that 3 of them will be flat-plate collectors?
48. Two dice are thrown 120 times. Find the average number of times in which the number on the first dice exceeds the number on the second dice.
49. It is known that the probability of an item produced by a certain machine will be defective is 0.05. If the produced items are sent to the market in packets of 20, find the number of packets containing at least, exactly and at most 2 defective items in a consignment of 1000 packets using (i) binomial distribution and (ii) Poisson approximation to binomial distribution.
50. If the probability is 0.2 that any one person will dislike the taste of new toothpaste, what is the probability that 5 of 18 randomly selected persons will dislike it.
51. If  $X$  is a Poisson variate such that  $2P(X=0) + P(X=2) = 2P(X=1)$ , find  $E(X)$ .
52. If  $X$  is a Poisson variate such that  $E(X^2) = 6$ , find  $E(X)$ .
53. If  $X$  is a Poisson variate such that  $P(X=0) = 0.5$ , find variance of  $X$ .
54. Patients arrive randomly and independently at a doctor's consulting room from 5 P.M at an average rate of one in 5 min. The waiting room can hold 12 persons. What is the probability that the room will be full, when the doctor arrives at 6 P.M.?
55. Find the mean and standard deviation of the distribution if the random variable is the number of students among 800 interviewed who do not like the food served at the university cafeteria, when the probability is 0.65 that any one of them does not like the food.
56. A particularly long traffic light on your morning commute is green 20% of the time that you approach it. Assume that each morning represents an independent trial.
- Over 5 mornings what is the probability that the light is green on exactly one day?
  - Over 20 mornings what is the probability that the light is green on atleast 4 days?
57. Determine the cumulative distribution function of a binomial random variable with  $n=3$  and  $p=0.25$ .

58. A multiple choice test contains 25 questions, each with 4 answers. Assume a student just guesses on each question.
- What is the probability that the student answers more than 20 questions correctly?
  - What is the probability that the student answers less than 5 questions correctly?
59. The number of messages sent to a computer bulletin board is a Poisson random variable with mean of 5 messages per hour.
- What is the probability that 5 messages are received in 1 hour?
  - What is the probability that 10 messages are received 1.5 hours?
  - What is the probability that less than 2 messages are received in one-half hour?
60. A binomial  $X$  satisfy the relation  $9P(X=4) = P(X=2)$  when  $n=6$ . Find the parameter  $p$  of the binomial distribution.
61. If 9 ships out of 10 arrive safely to a port, obtain the mean and standard deviation of a number of ships returning safely out of 150 ships.
62. The number of cracks in a section of inter state highway that are significant enough to require repair is assumed to follow a poisson distribution with mean of two cracks per mile. What is the probability that
- there are no cracks that require repair in 5 miles of highway?
  - at least one crack requires repair in  $\frac{1}{2}$  mile of highway?
63. The number of customers arriving at a cafeteria is a poisson RV with an average rate of 0.3 /minute. Find the probability that
- exactly 2 customers arrive in 10 minute span
  - 2 or more customers arrive in 10 minute span
64. The thickness of a flange on an aircraft component is uniformly distributed between 0.95 and 1.05 millimeters.
- Determine the cumulative distribution function of flange thickness.
  - Determine the proportion of flanges that exceeds 1.02 millimeters.
  - Determine the mean and variance of flange thickness.
65. Suppose the time it takes for a data collection operator to fill out an electronic form for a database is uniformly distributed between 1.5 and 2.2 minutes.
- What is the mean and variance of the time it takes an operator to fill out the form?
  - What is the probability that it will take less than 2 minutes to fill the form?
66. On a certain city transport route buses ply uniformly every 30 minutes between 6 am and 9 pm. If a person reaches a bus stop on this route at a random time during this period, what is the probability that he will have to wait for at least 10 minutes?
67. Suppose  $X$  has a uniform distribution over the interval  $[1.5, 5.5]$ . Determine (i) the mean and variance (ii)  $P(X < 2.5)$  (iii)  $P(X > 2 / X < 2.5)$  and (iv)  $P(X > 1.5 / X < 2.5)$
68. Suppose  $X$  has a continuous uniform distribution over the interval  $[-1, 1]$ . Determine (i) the mean and standard deviation (ii) the value of  $x$  such that  $P(-x < X < x) = 0.90$
69. If  $X$  is a random variable uniformly distributed in the interval  $(a, 9)$  and  $P(3 < X < 5) = 2/7$ . Find 'a' and compute  $P(|X-5| < 2)$ .
70. If  $X$  is a random variable uniformly distributed in the interval  $(-a, a)$ , determine a if (i)  $P(-1 < X < -2) = 0.75$ .

71. Determine the moment generating function of a uniformly distributed random variable in the interval  $(0,1)$ .
72. The length of a telephone conversation in a booth has been an exponential distribution and found on an average to be 5 minutes. Find the probability that a random call made from a booth (i) ends in less than 5 minutes. (ii) is between 5 and 10 minutes.
73. In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that a shower will last for 10 or more minutes.
74. Suppose  $X$  has an exponential distribution with mean equal to 10 .Determine :  
 a)  $P(X>10)$  b)  $P(X>20)$  c)  $P(X>30)$  d) Find  $x$  such that  $P(X<x) = 0.95$ .
75. The time to failure in hrs of fans in a personal computer can be modeled by an exponential distribution with  $\lambda = 0.0003$ .  
 (a)What proportion of fans will last at least 10,000 hrs ?  
 (b)What proportion of fans will last at most 7,000 hrs ?
76. The amount of time that a surveillance camera will run without having to be reset is a RV having exponential distribution with  $\beta = 50$  days. Find the probabilities that such a camera (a) will have to be reset in less than 20 days? (b) not have to be reset in atleast 60 days?
77. The amount of time that a watch will run without having to be reset is a random variable having an exponential distribution with mean 120 days. Find the probability that such a watch will  
 (i) have to be set in less than 24 days and  
 (ii) not have to be reset in at least 180 days.
78. The fill volume of an automated filling machine used for filling cans of carbonated beverage is normally distributed with mean of 12.4 fluid ounces and a standard deviation of 0.1 fluid ounce.  
 (i) What is the probability a fill volume is less than 12 fluid ounces?  
 (ii) If all cans less than 12.1 or greater than 12.6 ounces are scrapped, what proportion of cans is scrapped?
79. The time it takes a cell to divide (called mitosis) is normally distributed with an average time of one hour and a standard deviation of 5 minutes.  
 (i) What is the probability that a cell divides in less than 45 minutes?  
 (ii) What is the probability that it takes a cell more than 65 minutes to divide?  
 (iii) What is the time it takes approximately 99% of all cells to complete mitosis?
80. The time to microwave a bag of popcorn using the automatic setting can be treated as a RV having a normal distribution with standard deviation 10 seconds. If the probability is 0.8212 that the bag will take less than 282.5 seconds to pop, find the probabilities that  
 (i) that it will take longer than 258.3 seconds to pop and  
 (ii) that it will take on a value greater than 39.2 seconds.
81. Assume  $Z$  has a standard normal distribution. Use the standard normal table to find the value for  $z$  in each of the following cases.  
 (a)  $P(-1.24<Z<z) = 0.8$  (b)  $P(-z<Z<z) = 0.95$  (c)  $P(-z<Z<z) = 0.68$
82. The mean and standard deviation of a certain group of 1000 high school grades, that are normally distributed are 78% and 11% respectively. Find how many grades were above 90%?



83. In mathematics test, if we assume that your test scores( $X$ ) were approximately normally distributed with mean 76.8 and standard deviation 13.94.

- If a score below 60 represents a grade F(failure), approximately what percent of students failed in the test?
- If the cutoff for a grade of A is the lowest score of the top 15%, what is the cutoff mark.
- If the cutoff for a grade of C is the lowest score of the top 45%, what is the cutoff mark.
- How many marks must be added to the student scores so that only 10% fail (less than 60 be the failing grade)

84. Given is the joint probabilities of  $X$  and  $Y$ ,

| $X \backslash Y$ | 0    | 1    | 2    |
|------------------|------|------|------|
| 0                | 0.02 | 0.08 | 0.10 |
| 1                | 0.05 | 0.20 | 0.25 |
| 2                | 0.03 | 0.12 | 0.15 |

- Obtain the marginal distributions.
- Obtain the conditional distributions of  $X$ , given  $Y = 0$ .

85. Suppose the joint probability mass function of a bivariate random variable  $(X,Y)$  is given by:

$p(x,y) = 1/3$  for  $(0,1)$ ,  $(1,0)$ ,  $(2,0)$ ; 0, otherwise. Are  $X$  and  $Y$  independent?

86. The joint probability mass function pmf of  $X$  and  $Y$  is

| $x \backslash y$ | 0    | 1    | 2    |
|------------------|------|------|------|
| 0                | 0.1  | 0.04 | 0.02 |
| 1                | 0.08 | 0.20 | 0.06 |
| 2                | 0.06 | 0.14 | 0.30 |

Compute the marginal pmf of  $Y$ ,  $P[X \leq 1, Y \leq 1]$  and check if  $X$  and  $Y$  are independent.

87. The input to a binary communication system, denoted by a random variable  $X$ , takes on one of two values 0 or 1 with probabilities  $3/4$  and  $1/4$  respectively. Because of errors caused by noise in the system, the output  $Y$  differs from the input occasionally. The behaviour of the communication system is modeled by the conditional probabilities:  $P(Y=1/X=1) = 3/4$  and  $P(Y=0/X=0) = 7/8$ . Find  $P(Y=1)$ ,  $P(Y=0)$  and  $P(X=1/Y=1)$ .

88. The joint distribution of  $X_1$  and  $X_2$  is given by  $f(x_1, x_2) = (x_1 + x_2)/21$ ,  $x_1 = 1, 2$  and  $3$ ;  $x_2 = 1$  and  $2$ . Find the marginal distributions of  $X_1$  and  $X_2$ .

89. If  $X$  and  $Y$  are two random variables having joint density function

$f(x, y) = (1/8)(6 - x - y)$  for  $0 < x < 2$  and  $2 < y < 4$ . Find  $P(X < 1 \cap Y < 3)$

90. If  $X$  denotes the number of aces and  $Y$  the number of queens obtained when 2 cards are drawn at random (without replacement) from a deck of cards, obtain the joint probability distribution of  $(X,Y)$ .

91. The joint probability function of two discrete random variables  $X$  and  $Y$  is given by  $f(x,y) = c(2x+y)$ , where  $x$  and  $y$  can assume all integers such that  $0 \leq x \leq 2$  and  $0 \leq y \leq 3$  and  $f(x,y) = 0$  otherwise. Find the value of  $c$  and also find  $P(X \geq 1, Y \leq 2)$ .

92. Determine the value of  $c$  that makes the function  $f(x, y) = c(x + y)$  a joint probability mass function over the nine points with  $x = 1, 2, 3$  and  $y = 1, 2, 3$ . Also determine  $P(X=1, Y < 4)$ ,  $E(X)$ ,  $E(Y)$ ,  $V(X)$  and  $V(Y)$ .

93. Show that the following function satisfies the properties of a joint pmf and then evaluate  $P(X < 0.5)$ ,  $P(X > 0.25, Y < 4.5)$ ,  $E(X/Y=1)$  and  $P(X < 0.5, Y < 1.5)$ . Are  $X$  and  $Y$  independent.

| x    | y  | $f_{xy}(x,y)$ |
|------|----|---------------|
| -1   | -2 | 1/8           |
| -0.5 | -1 | 1/4           |
| 0.5  | 1  | 1/2           |
| 1    | 2  | 1/8           |

94. The joint pdf of the two dimensional random variable  $(X, Y)$  is given by,  $f(x,y) = (8/9)xy$ ,  $1 < x < y < 2$
- Find the marginal density functions of  $X$  and  $Y$ .
  - Find the conditional density function of  $Y$  given  $X$  and the conditional density function of  $X$  given  $Y$ .

95. If two random variables have the joint pdf  $f(x_1, x_2) = x_1x_2$ ,  $0 \leq x_1 \leq 1$ ,  $0 \leq x_2 \leq 2$ , find the probability that both random variables will take on values less than 1.

96. If the joint pdf of two random variables is given by  $f(x_1, x_2) = 6 \exp[-2x_1 - 3x_2]$  for  $x_1 \geq 0$ ,  $x_2 \geq 0$ , find  $P[1 \leq x_1 \leq 2, 2 \leq x_2 \leq 3]$ . Also find  $P[x_1 \leq 2, x_2 \geq 2]$ .

97. If the joint pdf of the random variable  $(X,Y)$  is given by  $f(x,y)=k(x^3y+xy^3)$ ,  $0 \leq x,y \leq 2$ . Find the value of  $k$  and also the conditional and marginal densities of  $X$  and  $Y$ .

98. A random variable  $X$  has a probability density function given by

$$f(x) = \begin{cases} 2e^{-2x}, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}$$

- Find the moment generating function and hence find the mean, variance and  $P(|X - \mu| > 1)$
- Use Chebyshev's inequality to obtain an upper bound on  $P(|X - \mu| > 1)$  and compare it with the result in (a).

99. Suppose that  $E(X) = 5$  and  $V(X) = 14$ . Compute the following quantities.

$$E(2X+1) \quad (ii) \quad E(X^2) \quad (iii) \quad V(2X+1)$$

100. If  $X$  and  $Y$  are independent normal variates with zero means and standard deviations 9 and 12 respectively and if  $X+2Y$  and  $kX-Y$  are uncorrelated, find  $k$ ?

101.  $X$  and  $Y$  are two random variables. It is found that  $X+Y$  has a variance of 15,  $X-Y$  has a variance of 11 and  $2X+Y$  has a variance of 29. Calculate the standard deviations of  $X$  and  $Y$ .

102. A fair die is tossed 720 times. Use Chebyshev's inequality to find a lowest bound for the probability of getting 100 to 140 sixes.

103. A random variable  $X$  is exponentially distributed with mean 1. Use Chebyshev's inequality to find  $P(-1 \leq X \leq 3)$ .

104. Two random variables  $X$  and  $Y$  have the joint density,  $f(x,y) = 2 - x - y$ ,  $0 \leq x \leq 1$ ,  $0 \leq y \leq 1$ ; 0, otherwise. Show that correlation of  $(X, Y) = -1/11$ .

105. The coefficient of correlation between  $X$  and  $Y$  is 0.48 and their covariance is 36. If the variance of  $X$  is 16, find the standard deviation of  $Y$ .

106. Consider a stochastic process defined on a finite sample space with three sample points. Its description is provided by the specifications of three sample functions  $X(t, s_1) = 3, X(t, s_2) = 3 \cos t, X(t, s_3) = 3 \sin t$ . Also given is the probability assignment  $P(S_i) = 1/3, i=1,2,3$ . Compute mean, second order moment and variance of the process. Also find the autocorrelation function of  $\{X(t)\}$ , given that any two RV's of the process are independent. Is the process a strict sense stationary or a wide sense stationary process?
107. Show that the process  $X(t) = A \cos pt + B \sin pt$ , where A and B are random variables is wide sense stationary if  $E(A) = E(B) = 0, E(A^2) = E(B^2)$  and  $E(AB) = 0$ .
108. Show that the random process  $X(t) = A \cos(\omega_0 t + \theta)$  is wide sense stationary, if A and  $\omega_0$  are constants and  $\theta$  is a uniformly distributed random variable in  $(0, 2\pi)$ .
109. When is a random process said to be evolutionary? Give an example of an evolutionary random process.
110. If  $X(t) = P + Qt$  where P and Q are independent random variables with  $E(P) = p$  and  $E(Q) = q, \text{Var}(P) = \sigma_1^2$  and  $\text{Var}(Q) = \sigma_2^2$ . Find the mean, auto correlation and auto covariance for  $\{X(t)\}$ . Is the process stationary?
111. If  $X(t) = \sin(\omega t + Y)$ , where Y is uniformly distributed in  $(0, 2\pi)$ , prove that  $\{X(t)\}$  is a wide sense stationary process.
112. Show that the process  $X(t) = A \cos pt + B \sin pt$ , where A and B are random variables is wide sense stationary if  $E(A) = E(B) = 0, E(A^2) = E(B^2)$  and  $E(AB) = 0$ .
113. If  $X(t) = X \sin t + Y \cos t$  for all t where X and Y are independent binary random variables, each of which assumes the values -1 and 2 with probabilities 1/3 and 2/3 respectively, prove that  $\{X(t)\}$  is wide sense stationary.
114. If  $X(t) = Y \cos t + Z \sin t$  for all t where Y and Z are independent binary random variables, each of which assumes the values -1 and 2 with probabilities 2/3 and 1/3 respectively, prove that  $\{X(t)\}$  is wide sense stationary.
115. If  $U(t) = X \cos t + Y \sin t$  and  $V(t) = Y \cos t + X \sin t$ , where X and Y are independent random variables such that  $E(X) = E(Y) = 0, E(X^2) = E(Y^2) = 1$ , show that  $\{U(t)\}$  and  $\{V(t)\}$  are individually stationary in the wide sense, but they are not jointly wide sense stationary.
116. If  $X(t) = 5 \cos(10t + p)$  and  $Y(t) = 20 \sin(10t + p)$  where p is a random variable uniformly distributed in  $(0, 2\pi)$ , prove that the process  $\{X(t)\}$  and  $\{Y(t)\}$  are jointly wide sense stationary.
117. A man is at an integral point on the X axis between the origin and the point x=3. He takes a unit step to the right with probability 1/3 or to the left with probability 2/3, unless he is at the origin, where he takes a step to the right to reach the point 1 or is at the point 3, where he takes a step to the left to reach the point 2. What is the probability that (i) he is at the point 1 after three walks? And (ii) he is at the point 3 after 4 walks.
118. A gambler has Rs.3. At each game, he loses Re 1 with probability 3/4, but wins Rs 2 with probability 1/4. He stops playing if he has lost his initial amount of Rs 3 or he has won at least Rs.3. Write down the transition probability matrix of the associated Markov chain.

119. The transition probability matrix of a markov chain with three states 0,1,2 is  $P = \begin{bmatrix} 3/4 & 1/4 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 3/4 & 1/4 \end{bmatrix}$  and the initial state distribution of the chain is  $p^{(0)} = (1/3 \quad 1/3 \quad 1/3)$ . Find  $P\{X_2 = 2\}$  and  $P\{X_3 = 1, X_2 = 2, X_1 = 1, X_0 = 2\}$ .

120. A Communication source can generate 1 of 3 possible messages 1,2 and 3. Assume that the generations can be described by a homogeneous Markov chain with the following transition probability matrix:

|   | <u>Current message</u> |     | <u>Next message</u> |
|---|------------------------|-----|---------------------|
|   | 1                      | 2   | 3                   |
| 1 | 0.5                    | 0.3 | 0.2                 |
| 2 | 0.4                    | 0.2 | 0.4                 |
| 3 | 0.3                    | 0.3 | 0.4                 |

and the initial state probability distribution  $p^{(0)} = (0.3 \quad 0.3 \quad 0.4)$ . Find  $p^{(3)}$ .

121. A message transmission system is found to be Markovian with the transition probability of current message to the next message as given by the matrix below. The initial probabilities of the states are  $p^{(0)} = (0.4 \quad 0.3 \quad 0.3)$ . Find the probability of the next message.

$$P = \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.2 & 0.7 \\ 0.6 & 0.3 & 0.1 \end{bmatrix}$$

122. If the customers arrive at a bank according to a Poisson process with mean rate of 2 per minute, find the probability that during a 1-min interval, no customer arrives.
123. On the average, a submarine on patrol sights 6 enemy ships per hour. Assuming that the number of ships sighted in a given length of time is a Poisson variate, find the probability of sighting (i) six ships in the next half-an-hour, (ii) 4 ships in the next 2 hours, (iii) atleast one ship in the next 15 min and (iv) atleast two ships in the next 20 min.

124. Find the mean and variance of stationary process  $\{X(t)\}$  whose auto correlation function is

$$R(\tau) = \frac{25\tau^2 + 36}{6.25\tau^2 + 4}$$

125. Three boys A, B, C are throwing a ball to each other. A always throws the ball to B, B always throws to C, C is just as likely to throw the ball to B as to A. Find the transition matrix.
126. Message arrive at a telegraph office in accordance with the laws of a Poisson process with a mean rate of 3 messages per hour. What is the probability that no message will have arrived during the morning hours, i.e between 8 A.M and 12 noon.

127. The transition probability matrix of a Markov chain with three states 0,1,2 is  $P = \begin{bmatrix} 3/4 & 1/4 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 3/4 & 1/4 \end{bmatrix}$  and the initial state distribution of the chain is  $p^{(0)} = (1/3 \quad 1/3 \quad 1/3)$ . Find  $P\{X_2 = 2\}$  and  $P\{X_3 = 1, X_2 = 2, X_1 = 1, X_0 = 2\}$ .

128. The transition probability matrix of a Markov chain  $\{X_n\}$   $n=0,1,2,\dots$  having 3 states 1, 2 and 3 is
- $$P = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix}$$
- and the initial probability distribution  $P^{(0)} = [0.7, 0.2, 0.1]$ . Find  $P(X_2 = 3)$ ,  $P(X_3 = 2, X_2 = 3, X_1 = 3, X_0 = 2)$ .
129. Prove that the difference of two independent Poisson Processes is not a Poisson Process.
130. A random process  $\{X(t), t \in T\}$  has the probability distribution
- $$P(X(t) = n) = \begin{cases} \frac{(at)^{n-1}}{(1+at)^{n+1}}, & n = 1, 2, 3, \dots \\ \frac{at}{1+at}, & n = 0 \end{cases}$$
- . Show that the process is evolutionary.
131. Define autocorrelation function & prove that for a WSS process  $\{X(t)\}$   $R_{xx}(\tau) = R_{xx}(-\tau)$ .
132. The auto correlation function for a stationary process is given by  $R_{xx}(\tau) = 9 + 2e^{-|\tau|}$ . Find the mean and variance of  $X(t)$ .
133. The autocorrelation function for a stationary process is given by  $R_{xx}(\tau) = 25 + \frac{4}{1+6\tau^2}$ . Find the mean and variance of  $X(t)$ .
134. Given a random variable  $Y$  with characteristic function  $\phi(w) = E[e^{iwY}]$  and a random process defined by  $X(t) = \cos(\lambda t + Y)$ , Show that  $X(t)$  is stationary in the wide sense if  $\phi(1) = \phi(2) = 0$ .
135. A random process has sample functions of the form  $X(t) = A \cos(\omega t + \theta)$  where  $\omega$  is a constant,  $A$  and  $\theta$  are independent random variables and  $\theta$  is uniformly distributed between 0 and  $2\pi$ .
- Is  $X(t)$  a wide-sense stationary process?
  - Is  $X(t)$  a mean ergodic process?
136. Show that the sum of two independent Poisson process is a Poisson process and the difference of two independent Poisson process is not a Poisson process.
137. The power spectral density of a random process is given by  $S_{xx}(\omega) = \begin{cases} \pi & |\omega| < 1 \\ 0 & \text{otherwise} \end{cases}$ . Find its autocorrelation function.
138. A student's study habits are as follows: If he studies one night, he is 60% sure not to study the next night; on the other hand if he study one night, he is 80% sure to study the next night. In long run how often does he study?
139. A hospital receives an average of 3 emergency calls in a 10 minute interval. What is the probability that there are at most 3 emergency calls in a 10 minute interval?
140. A random process is defined by  $X(t) = K \cos \omega t$ ,  $t \geq 0$ , where  $\omega$  is a constant and  $K$  is uniformly distributed between 0 and 2. Determine the mean, autocorrelation function and the autocovariance function of  $X(t)$ .

141. Given that the autocorrelation function for a stationary ergodic process with no periodic components is  $R_{xx}(\tau) = 16 + \frac{28}{\tau^2 + 1}$ . Find the mean and variance of the process  $\{X(t)\}$ .
142. Find the power spectral density of the random process whose auto correlation function is  $R(\tau) = e^{-\alpha|\tau|} \cos \beta\tau$
143. Determine the mean and variance of the Poisson process.
144. A software engineer goes to his work-place every day by motor bike or by car. He never goes by bike on two consecutive days; but if he goes by car on a day then he is of equally likely to go by car or by bike on the next day. Find the transition matrix for the chain of the mode of transport he uses. If car is used on the first day of a week, find the probability that (i) bike is used (ii) car is used, on the fifth day.
145. A radioactive source emits particles at a rate of 5 per minute according to a Poisson process. Each particle emitted has a probability 0.6 of being recorded. Find the probability that 10 particles are recorded in a 4 minute period.
146. A man either drives a car or catches a train to go to office each day. He never goes two days in a row by train, but if he drives one day, then the next day he is just as likely to drive again he is to travel by train. Now suppose that on the first day of the week, the man tossed a fair die and drove to work if and only if a 6 appeared. Find (i) The probability that he takes a train on the third day (ii) The probability that he drives to work in the long run.
147. If the transition probability matrix of a Markov Chain is  $\begin{bmatrix} 0 & 1 \\ 0.5 & 0.5 \end{bmatrix}$  find the steady state distribution of the chain.
148. Find the nature of the states of the Markovian chain with the transition probability matrix  $\begin{bmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$ .
149. Three boys A, B and C are throwing a ball to each other. A always throws the ball to B and B always throw the ball to C; but C is just as likely to throw the ball to B as to A. Show that the process is Markovian. Find the transition matrix and classify the states.
150. Consider two random processes  $X(t) = 3\cos(\omega t + \theta)$  and  $Y(t) = 2\cos(\omega t + \theta - \frac{\pi}{2})$  where  $\theta$  is a random variable uniformly distributed in  $(0, 2\pi)$ . P.T.  $\sqrt{R_{xx}(0) R_{yy}(0)} \geq |R_{xy}(\tau)|$ .